Aalborg Universitet

AALBORG UNIVERSITY

HearWalk: Co-designing and Building a Sound Feedback System for Hemiparetic Gait Training

Kantan, Prithvi Ravi; Dahl, Sofia; Rovsing Møller Jørgensen, Helle; Lanther, Maria Barbro; Spaich, Erika G.

Creative Commons License
Unspecified

Publication date:
2023

Document Version
Publisher's PDF, also known as Version of record

Link to publication from Aalborg University

Citation for published version (APA):
Kantan, P. R., Dahl, S., Rovsing Møller Jørgensen, H., Lanther, M. B., & Spaich, E. G. (2023). *HearWalk: Co-designing and Building a Sound Feedback System for Hemiparetic Gait Training*. Abstract from Regionshospital Nordjyllands Forskningssymposium 2023, Hjørring, Denmark.

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal -

Take down policy
If you believe that this document breaches copyright please contact us at vbn@aub.aau.dk providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from vbn.aau.dk on: 11 07, 2026





HearWalk: Co-designing and Building a Sound Feedback System for Hemiparetic Gait Training

Prithvi Ravi Kantan[1] • Sofia Dahl[1] • Helle Rovsing Jørgensen[2] • Maria Barbro Lanther[1] • Erika G. Spaich[3]

[1] Department of Architecture, Design and Media Technology, Aalborg University - Copenhagen, Denmark
[2] Neuroenhed Nord, Regionhospital Nordjylland, Brønderslev, Denmark
[3] Center for Neurotechnology and Rehabilitation, Department of Health Science and Technology, Aalborg University - Aalborg, Denmark

DEMO VIDEOS

Background

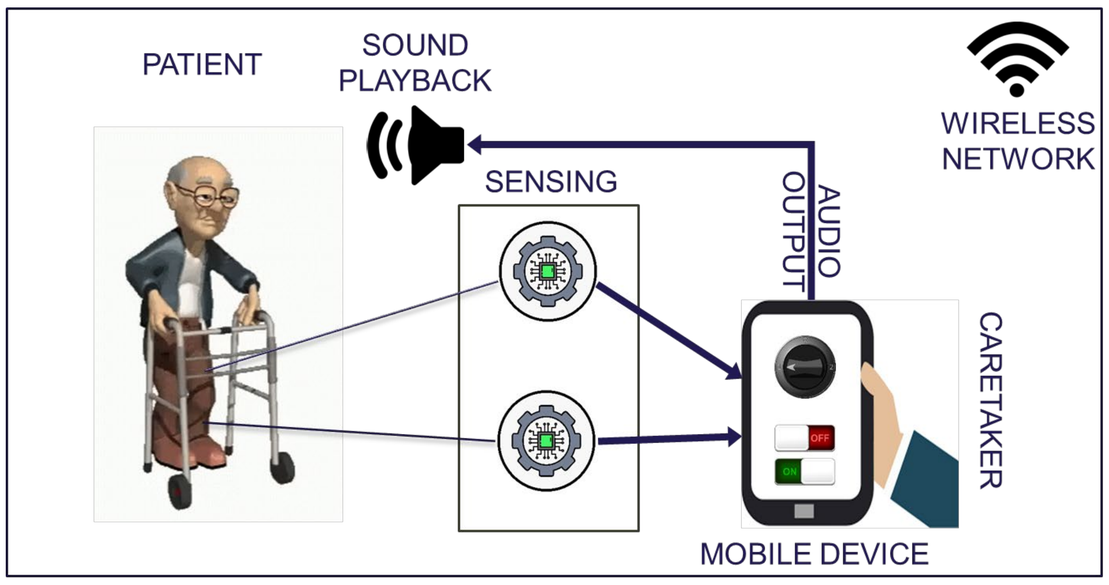
Sound feedback technology has been increasingly explored as a tool to enhance motor learning during gait rehabilitation.

Hemiparetic patients typically exhibit **intra-group variability**, so it is very important for assistive technological systems to be **patient-tailored** and cater to the diversity of individual needs.

It is recommended that **patients and professionals are systematically and meaningfully involved** in technology development processes so as to **enhance the usability and user acceptance** of the technology.

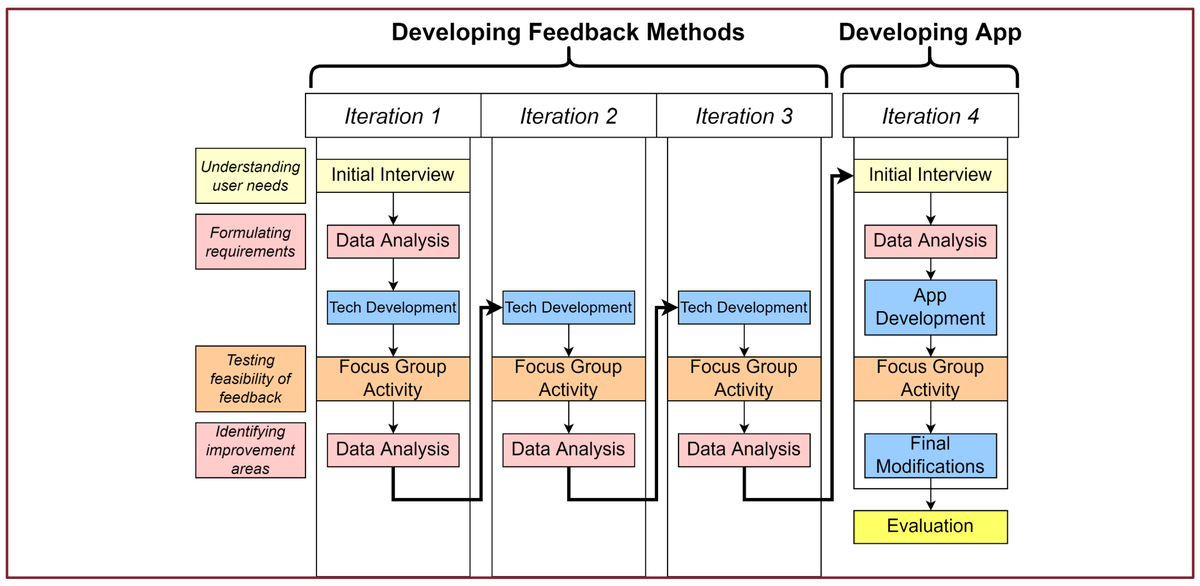
Objective

To collaborate with physiotherapists on developing and testing the feasibility of a low-cost sound feedback system with multiple feedback designs compatible with conventional gait training protocols.



Use-case Scenarios

Exercise Name	Focus	Overview	Patient Goal
Overground Walk – **Wading Feedback**	Symmetry Swing Phase	Sounds like walking through water	Identical sound on left / right
Single-sided stand – **Music Feedback**	Knee Extension Control	Music interrupted by noise if knee flexes or hyperextends	Maintain full knee extension without hyperextending
Overground Walk – **Step Length Feedback**	Paretic Step Length	Reward 'PLING' sound if desired step length achieved	Achieve reward sound during every paretic limb step



Methods

Our research employed a **user-centered approach**, encapsulated in four iterative cycles of design, development, and evaluation.

Key to this process was the **active involvement of a focus group** of therapists, who collaboratively delineated clinical use-case scenarios, defined general patients' needs and abilities, and proposed sound feedback ideas.

Following each development phase, the resulting strategies underwent rigorous scrutiny by the **focus group in an interview and testing setting**. To assess practicality and usability of the sound feedback, we executed **feasibility studies involving patient-therapist pairs**.

Subsequent iterations were shaped by their expert insights, leading to the **final prototype system** comprising a portable wireless system made up of inexpensive motion sensors, a mobile app, and a sound playback system.

Takeaways and Perspectives

We defined, designed, and developed three **concrete use case scenarios (see demo videos)**, which were found to be easily integrable with conventional rehabilitation protocols and of potential therapeutic benefit to several patient types.

Patients exhibited a **large amount of variability in terms of physical and cognitive abilities, general auditory comprehension, and preferences**, which supports the notion that the feedback may not be universally applicable, and that the system needs to be adjustable to the individual patient.

Further research and development will be necessary to develop **use-cases for specific patient subgroups and movement impairments**, which is the subject of future work.

Overall, despite the expensive and time-consuming nature of user-centered design methodologies, this project adds to a growing body of evidence that **clinical stakeholder perspectives are crucial to the effective development and implementation** of rehabilitation technology at large.

BETA.HEALTH®
NordForsk
REGION NORDJYLLAND – i gode hænder